Tristin Xie

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Education

University of California San Diego

M.S. in Computer Science — *Sep. 2023 - Present*

University of California San Diego — *CS GPA: 3.722 / 4.0*

Double Major: B.S. in Computer Science and Applied Mathematics · Minor in Business — *Sep. 2019 - Sep. 2023*

- **Tau Beta Pi:** Oldest engineering honor society with invitations only sent to the **top 1/8th of the Junior class and top 1/5th of the Senior class**
- **Graduate Courses:** Computer Vision I · Operating Systems · Introduction to Robotics · Networked Services · Deep Learning · Machine Learning for Music · ML: Learning Algorithms · Statistical Natural Language Processing · Compiler Construction · Computability and Complexity
- **Undergraduate Courses:** Computer Vision I and II · Deep Learning · Computer Graphics · Recommender Systems · Linear and Nonlinear Programming · Optimization for Data Science · Math Statistics I and II · Probability · Linear Algebra · Software Engineering · Advanced Data Structures and Algorithms · Operating Systems · Compilers · Computer Architecture · Programming Languages · Systems Programming · Real Analysis · Vector Calculus · Differential Equations

Experience

Advanced Robotics and Controls Lab (ARClab) at UC San Diego — *San Diego, CA*

Research Assistant — *Mar. 2023 - Present*

- Wrote a **particle filter** with **COLMAP**, **ORB-SLAM3**, **DPVO** motion models in **PyTorch** for robust camera-to-robot pose estimation
- Created a data collection pipeline for the **Franka Emika Panda** robot and a **Microsoft Kinect** through **Robot Operating System (ROS)**
- Evaluated estimated camera-to-robot pose segmentation masks with **Intersection over Union** using **Segment Anything 2** as ground truth
- Experimented with **statistical filtering techniques** such as z-score, Kalman, and softmax confidence to reduce erratic predictions

Amazon Web Services — *Seattle, WA*

Software Development Engineer Intern — *Jun. 2023 - Sep. 2023*

- Overhauled **autocomplete** from a static frontend approach to a dynamic backend approach reflecting real data in **Elasticsearch**
- Tested implementation by writing new **unit, integration, and canary tests** and created **AWS CloudWatch alarms, metrics, and dashboards**
- **Improved search results by 28%** for exact keyword match queries of a major category, diagnosed by writing a **multithreaded Python** script

Software Development Engineer Intern — *Jun. 2022 - Sep. 2022*

- Implemented and tested an **AWS DynamoDB** schema and **AWS Lambda** API using **Go** to support Word-like notes for **10,000+ AWS partners**
- Defined cloud infrastructure using **AWS Cloud Development Kit** with **TypeScript** and deployed **CloudFormation** stacks for my application

UC San Diego Computer Science and Engineering — *San Diego, CA*

Instructional Assistant — *Jul. 2021 - Mar. 2023*

- Tutored preparatory and lower division courses on **algorithms, recursion, state machines, Pygame, Python, and image manipulation**
- Tutored upper division course with assigned **C++** programming assignments and conceptual problems for a combined total of **100+ hours**
- Tested **30+** advanced data structures assignments including **self-balancing binary search trees, tries, hash maps, bloom filters, and more**

UC San Diego Design Lab — *San Diego, CA*

Research Assistant · Research Ethics Certified By ReCODE health — *Sep. 2020 - Nov. 2020*

- Cooperated with **Ford Motor Company** to explore **autonomous car implications** and functionalities for the driver
- Conducted research on the technical team and presented findings to **45+ fellow research assistants** and professors

Publications

CtRNet-X: Camera-to-Robot Pose Estimation in Real-world Conditions Using a Single Camera

Jingpei Lu[†], Zekai Liang[†], **Tristin Xie**, Florian Ritcher, Shan Lin, Sainan Liu, Michael C. Yip — *[†] Equal contribution*

2025 IEEE International Conference on Robotics and Automation (ICRA), pp. 1914-1920

Leadership

Association for Computing Machinery at UC San Diego — *San Diego, CA*

Vice President Projects — *May 2022 - May 2023*

- Oversaw the processing of **250+ applicants** and hosted a project showcase event with **90+ attendance** for the quarterly ACM Projects program

Vice President Finance — *May 2021 - May 2022*

- Directed the finance team to create financial documents to **obtain funding** for **200+ events** from school departments and student government
- Managed and budgeted **thousands of dollars** of department funds, sponsorship money, and other miscellaneous University allocated funds
- Contacted **5+ vendors** and created a pipeline for receiving invoices which allows **direct payment by University instead of reimbursement**

Projects

ACM Teammate Finder: Web App To Find Teammates Easily | GitHub *Oct. 2020 - Jan. 2021*

- Implemented **Material Design** webpages with **React, HTML/CSS, and React Material Web Components**
- Developed **CRUD functions** for back-end using **Express** to fetch and post updates to the **Firestore** database

EnviroSense: Internet of Things Pollution Sensor | Devpost | GitHub *Oct. 2019*

- **SDHacks WINNER (3rd):** *Best IoT that Incorporates Multiple Nodes (team of 4)*
- Built a sensor node prototype which measures pollutant and environmental data from the ambient environment

PictureTranslate: Real-time Computer Vision Object Translator | GitHub *Aug. 2019*

- Pair programmed and trained **Haar Cascades** on 4 different smartphones, 4 denominations of money, and more
- Displayed video feed and paused with translation of the detected object to **100+ supported languages** using the **Google Translate API**

Skills

Languages	Python · C/C++ · Go · JavaScript/TypeScript · Haskell · HTML/CSS · Sass · ARM Assembly · Bash · Java · LaTeX
Frameworks and Tools	NumPy · PyTorch · AWS · Git · Docker · React · Node.js · Express.js · Tailwind CSS · Firebase · OpenCV · Jest · GDB
Software	Robot Operating System (ROS) · Vim · Visual Studio Code · Visual Studio · IntelliJ IDEA Ultimate · Adobe Illustrator
Foreign Languages	Fluent in Mandarin Chinese